

And, we have that:

$$\begin{aligned} A_1 &= \mathbb{E}[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} \Sigma \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} \tilde{\mathbf{x}}_j] \\ &= \lambda^2 \mathbb{E}[y_i \tilde{\mathbf{x}}_i^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} \Sigma \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} y_j \tilde{\mathbf{x}}_j] \\ &= \lambda^2 \boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta \end{aligned}$$

And:

$$\begin{aligned} A_2 &= \frac{1}{N(1+\delta_S)} \mathbb{E}[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} \tilde{\mathbf{x}}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} \Sigma \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} \tilde{\mathbf{x}}_j] \\ &= \frac{1}{N(1+\delta_S)} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \mathbb{E}[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_{i,j}} \tilde{\mathbf{x}}_j] \\ &= \frac{\lambda^2}{N(1+\delta_S)} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \end{aligned}$$

And, we can easily observe that:

$$A_3 = A_2, \quad A_4 = \mathcal{O}(N^{-1})$$

Thus:

$$\frac{1}{N^2} \sum_{i \neq j} \mathbb{E}[q_i q_j \tilde{y}_i \tilde{y}_j \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_j^\top \mathbf{Q} \mathbf{x}] = \frac{\lambda^2(1-\pi)^2}{(1+\delta_S)^2} \left(\boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta - \frac{2}{N(1+\delta_S)} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right)$$

- And for $i \in \{1, \dots, m\}$:

$$\begin{aligned} \mathbb{E}[q_i \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_i^\top \mathbf{Q} \mathbf{x}] &= \frac{1}{(1+\delta_S)^2} \mathbb{E}[q_i \tilde{\mathbf{x}}_i^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_i} \Sigma \mathbf{Q}_{-\tilde{\mathbf{x}}_i} \tilde{\mathbf{x}}_i] \\ &= \frac{\alpha}{(1+\delta_S)^2} \mathbb{E}[\tilde{\mathbf{x}}_i^\top \mathbf{Q}_{-\tilde{\mathbf{x}}_i} \Sigma \mathbf{Q}_{-\tilde{\mathbf{x}}_i} \tilde{\mathbf{x}}_i] \\ &= \frac{\alpha}{(1+\delta_S)^2} \text{Tr}(\mathbb{E}[\tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top] \mathbb{E}[\mathbf{Q}_{-\tilde{\mathbf{x}}_i} \Sigma \mathbf{Q}_{-\tilde{\mathbf{x}}_i}]) \\ &= \frac{\alpha}{(1+\delta_S)^2} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \end{aligned}$$

Hence, by grouping the terms, the second sum gives us:

$$\frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^m q_i \tilde{y}_i \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x} \right)^2 \right] \tag{17}$$

$$= \frac{\lambda^2(1-\pi)^2}{(1+\delta_S)^2} \left(\boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta - \frac{2}{N(1+\delta_S)} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) + \frac{\alpha(1-\pi)}{N(1+\delta_S)^2} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \tag{18}$$

Third sum: Let us now compute the remaining term in the sum that is given by:

$$\frac{2}{N^2} \sum_{i=1}^n \sum_{j=1}^m \mathbb{E}[y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} q_j \tilde{y}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x}]$$

Let $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, m\}$, we have that:

$$\begin{aligned} \mathbb{E}[y_i q_j \tilde{y}_j \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \mathbf{x}_j^\top \mathbf{Q} \tilde{\mathbf{x}}_j] &= \mathbb{E}[y_i q_j \tilde{y}_j \mathbf{x}_i^\top \mathbf{Q} \Sigma \mathbf{Q} \tilde{\mathbf{x}}_j] \\ &= \frac{1}{(1+\delta)(1+\delta_S)} \mathbb{E}[y_i q_j \tilde{y}_j \mathbf{x}_i^\top \mathbf{Q}_{-\mathbf{x}_i} \Sigma \mathbf{Q}_{-\tilde{\mathbf{x}}_j} \tilde{\mathbf{x}}_j] \\ &= \frac{1}{(1+\delta)(1+\delta_S)} \mathbb{E} \left[y_i q_j \tilde{y}_j \mathbf{x}_i^\top \left(\mathbf{Q}_{-ij} - \frac{\mathbf{Q}_{-ij} q_j \tilde{\mathbf{x}}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q}_{-ij}}{N(1+\delta_S)} \right) \Sigma \left(\mathbf{Q}_{-ij} - \frac{\mathbf{Q}_{-ij} \mathbf{x}_i \mathbf{x}_i^\top \mathbf{Q}_{-ij}}{N(1+\delta)} \right) \tilde{\mathbf{x}}_j \right] \\ &= \frac{1}{(1+\delta)(1+\delta_S)} (A_1 - A_2 - A_3 + A_4) \end{aligned}$$

We have that:

$$\begin{aligned} A_1 &= \mathbb{E}[y_i q_j \tilde{y}_j \mathbf{x}_i^\top \mathbf{Q}_{-ij} \Sigma \mathbf{Q}_{-ij} \tilde{\mathbf{x}}_j] = \lambda \mathbb{E}[y_i \mathbf{x}_i^\top \mathbf{Q}_{-ij} \Sigma \mathbf{Q}_{-ij} \tilde{y}_j \tilde{\mathbf{x}}_j] \\ &= \lambda \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta \end{aligned}$$

And:

$$\begin{aligned} A_2 &= \frac{1}{N(1+\delta)} \mathbb{E}[y_i q_j \tilde{y}_j \mathbf{x}_i^\top \mathbf{Q}_{-ij} \Sigma \mathbf{Q}_{-ij} \mathbf{x}_i \mathbf{x}_i^\top \mathbf{Q}_{-ij} \tilde{\mathbf{x}}_j] \\ &= \frac{\lambda}{N(1+\delta)} \mathbb{E}[y_i \mathbf{x}_i^\top \mathbf{Q}_{-ij} \Sigma \mathbf{Q}_{-ij} \mathbf{x}_i \mathbf{x}_i^\top \mathbf{Q}_{-ij} y_j \tilde{\mathbf{x}}_j] \\ &= \frac{\lambda}{N(1+\delta)} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \end{aligned}$$

And also:

$$\begin{aligned} A_3 &= \frac{1}{N(1+\delta_S)} \mathbb{E}[y_i q_j \tilde{y}_j \mathbf{x}_i^\top \mathbf{Q}_{-ij} \tilde{\mathbf{x}}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q}_{-ij} \Sigma \mathbf{Q}_{-ij} \tilde{\mathbf{x}}_j] \\ &= \frac{\lambda}{N(1+\delta_S)} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \end{aligned}$$

Hence:

$$\frac{2}{N^2} \sum_{i=1}^n \sum_{j=1}^m \mathbb{E}[y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} q_j \tilde{y}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x}] \quad (19)$$

$$= \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)} \left(\boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta - \frac{1}{N} \text{Tr} \left(\left(\frac{\Sigma}{1+\delta} + \frac{\Sigma_\beta}{1+\delta_S} \right) \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \right) \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \quad (20)$$

Grouping all the sums: Denote by : $T_1 = \frac{1}{N} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])$, then: $T_2 = \frac{1}{N} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])$. Now let us group the terms in T in the three sums, and those that do not depend on T . We get that:

$$\begin{aligned} \mathbb{E}[(\mathbf{w}^\top \mathbf{x})^2] &= \frac{\pi^2}{(1+\delta)^2} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} + \frac{\lambda^2(1-\pi)^2}{(1+\delta_S)^2} \boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta + \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta \\ &+ T_1 \left(\frac{\pi}{(1+\delta)^2} - \frac{2\pi^2}{(1+\delta)^3} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} - \frac{2\lambda\pi(1-\pi)}{(1+\delta)^2(1+\delta_S)} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \\ &+ T_2 \left(\frac{\alpha(1-\pi)}{(1+\delta_S)^2} - \frac{2\lambda^2(1-\pi)^2}{(1+\delta_S)^3} \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta - \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)^2} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \\ &= \frac{\pi^2}{(1+\delta)^2} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} + \frac{\lambda^2(1-\pi)^2}{(1+\delta_S)^2} \boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta + \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta \\ &+ \frac{\pi T_1}{(1+\delta)^2} \left(1 - \frac{2\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} - \frac{2\lambda(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \\ &+ \frac{(1-\pi)T_2}{(1+\delta_S)^2} \left(\alpha - \frac{2\lambda^2(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta - \frac{2\lambda\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \end{aligned}$$

This leads to the following theorem:

Theorem B.1 (Gaussianity of the General model). *Let $\mathbf{w}_q$ be the Mixed classifier as defined in equation 6 and suppose that Assumption 4.1 holds. The decision function $\mathbf{w}_q^\top \mathbf{x}$, on some test sample $\mathbf{x} \in \mathcal{C}_a$ independent of $\mathbf{X}$, satisfies:*

$$\mathbf{w}_q^\top \mathbf{x} \xrightarrow{\mathcal{D}} \mathcal{N}((-1)^a m_q, \nu_q - m_q^2),$$

where:

$$\begin{aligned}
m_q &= \left(\frac{\pi}{1+\delta} \boldsymbol{\mu}^\top + \frac{\lambda(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \right) \bar{\mathbf{Q}} \boldsymbol{\mu}, \\
\nu_q &= \frac{\pi^2}{(1+\delta)^2} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} + \frac{\lambda^2(1-\pi)^2}{(1+\delta_S)^2} \boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta + \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta \\
&\quad + \frac{\pi T_1}{(1+\delta)^2} \left(1 - \frac{2\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} - \frac{2\lambda(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \\
&\quad + \frac{(1-\pi)T_2}{(1+\delta_S)^2} \left(\alpha - \frac{2\lambda^2(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta - \frac{2\lambda\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right).
\end{aligned}$$

where:

$$T_1 = \frac{1}{N} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]), \quad T_2 = \frac{1}{N} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]), \quad \lambda = \phi(1-\varepsilon) - \rho\varepsilon$$